

Cyclistic Bike-Share Case Study

Google Data Analytics Capstone Project

Author: Abhirami Ananthakumar

Tools: R • Tableau • Excel • GitHub

Dataset: Divvy Bike Trips (July 2025 – June 2026)

Project Objective

Analyze 12 months of Cyclistic bike-share data to understand the behavioral differences between annual members and casual riders and recommend strategies to convert casual riders into annual members.

Business Problem

Cyclistic wants to increase the number of annual memberships because members are more profitable than casual riders. The marketing team needs data-driven insights into how rider behavior differs between the two customer segments.

Business Task

Identify patterns in ride behavior by comparing casual riders and annual members and provide actionable business recommendations that support membership growth.

Dataset

Include:

- Source: Divvy Bike Share (Cyclistic)
- Period: July 2025 – June 2026
- Total Records: 5.93 Million
- File Size: 1.42 GB
- Data Type: CSV

Tools Used

Tool	Purpose
R (tidyverse, lubridate, janitor)	Data cleaning, processing, and analysis
Tableau	Dashboard development and data visualization

Google Docs	Project documentation
Google Slides	Project presentation
GitHub	Portfolio hosting and version control

Methodology

The Google Data Analytics phases:

1. Ask
2. Prepare
3. Process
4. Analyze
5. Share
6. Act

Key Findings

- Annual members completed significantly more rides than casual riders.
- Casual riders had a much longer average ride duration.
- Members mainly used bikes for weekday commuting.
- Casual riders rode more frequently during weekends.
- Ride demand peaked during the summer months for both rider types.
- Electric bikes were popular among both annual members and casual riders.

Business Recommendations

1. Offer weekend membership discounts targeting casual riders.
2. Launch seasonal promotions during summer when casual usage is highest.
3. Create commuter-focused membership campaigns highlighting savings for frequent riders.
4. Provide loyalty rewards for casual riders after completing a certain number of rides.
5. Promote electric bike membership benefits through targeted marketing.

Business Impact

This analysis provides actionable insights into customer behavior that can help Cyclistic increase annual memberships. The recommendations are based on observed riding patterns and can support marketing decisions aimed at improving customer conversion and long-term profitability.

Skills Demonstrated

- Data Cleaning in R
- Data Transformation
- Data Visualization
- Exploratory Data Analysis (EDA)
- Business Analysis
- Data Storytelling
- Dashboard Design
- Business Recommendation Development
- Documentation

Conclusion

The analysis revealed clear behavioral differences between annual members and casual riders. Annual members primarily use bicycles for regular commuting, while casual riders mainly use them for leisure activities during weekends and warmer months. These insights provide a strong foundation for marketing strategies designed to convert casual riders into annual members.